

	$= 2.1 \times 10^7 \text{ m s}^{-1}$
2(a)	Evaporation occurs at any temperature while boiling occurs at boiling point only.
2(b)	Work done against atmosphere $= (1.05 \times 10^5) (1.69)$ $= 1.77 \times 10^5 \text{ J}$
2(c)	For 1.00 kg of water, $Q = ml$ $= (1.00) (2.30 \times 10^6)$ $= 2.30 \times 10^6 \text{ J}$ $\Delta U = Q + W$ $= 2.30 \times 10^6 - 1.77 \times 10^5$ $= 2.12 \times 10^6 \text{ J}$ Number of water molecules in 1.00 kg $= \frac{1.00}{0.0180} (6.02 \times 10^{23})$ $= 3.34 \times 10^{25}$ Average increase $= \frac{2.12 \times 10^6}{3.34 \times 10^{25}}$ $= 6.30 \times 10^{-20} \text{ J}$